Project Work, part 1

General All project work in IND320 will result in personal hand-ins and online apps.

A Jupyter Notebook run locally on your computer (later with access to online and local databases). This will be your basic development and documentation platform.

Must include a brief description of AI usage. Must include a 300-500-word log describing the compulsory work (including both Jupyter Notebook and Streamlit experience). Must include links to your public GitHub repository and Streamlit app (see below) for the compulsory work.

Document headings should be clear and usable for navigation during development. All code blocks must include enough comments to be understandable and reproducible if someone inherits your project.

All code blocks must be run before an export to PDF so the messages and plots are shown. In addition, add the .ipynb file to the GitHub repository where you have your Streamlit project. A Streamlit app running at https://[yourproject].streamlit.app/.

This will be an online version of the project, accessing data that has been exported to CSV format (later, also an online database).

The code, hosted on GitHub, must include relevant comments from the Jupyter Notebook and further comments regarding Streamlit usage.

There are four parts in the project work, building on each other and resulting in a final portfolio and app to be presented at the end of the semester.

Co-operation is applauded, and the use of AI tools is encouraged.

Jupyter Notebook Read the reservoirs.csv file using Pandas (located in D2Dbook/data).

Rename headers to English and understandable.

Print its contents in a relevant way.

Plot each column separately.

Plot all columns together. Consider how to make this natural, given that the scales are different.

Remember to fill in the log and AI mentioned in the General section above.

Importing libaries:

In [13]:
```python
import pandas as pd
import matplotlib.pyplot as plt
```

Github rep: https://github.com/IsmailNabi/IND320-prosjekt

Streamlit app: https://ismail.streamlit.app

2.1 Read the supplied data

In [14]:
```python
df = pd.read_csv('/Users/ismailnabi/Desktop/Indøk/IND320_Prosjekt/data/reservoirs.csv')
df.head()
```

Out[14]:

	dato_Id	omrType	omrnr	iso_aar	iso_uke	fyllingsgrad	kapasitet_TWh	fylling_TWh	neste_Publiseringsdato	fyllingsgrad_forrige_uke	endring_fyllingsgrad
0	1995-09-03	EL	4	1995	35	0.944721	21.079208	19.913979	0001-01-01T00:00:00	0.936888	0.007833
1	2020-11-15	EL	1	2020	46	0.974361	6.003264	5.849344	2020-11-25T13:00:00	0.997406	-0.023046
2	2011-08-28	EL	4	2011	34	0.786499	21.079208	16.578772	0001-01-01T00:00:00	0.787193	-0.000694
3	1998-07-05	EL	3	1998	27	0.793969	8.920932	7.082947	0001-01-01T00:00:00	0.733812	0.060157
4	2011-03-27	EL	4	2011	12	0.300820	21.079208	6.341054	0001-01-01T00:00:00	0.311122	-0.010301

2.2 Print useful information

In [15]:
```python
#Column mapping to rename columns for better readability
column_mapping = {
    'dato_Id': 'date',
    'omrType': 'area_type',
    'omrnr': 'area_number',
    'iso_aar': 'iso_year',
    'iso_uke': 'iso_week',
    'fyllingsgrad': 'filling_degree',
    'kapasitet_TWh': 'capacity_twh',
    'fylling_TWh': 'filling_twh',
    'neste_Publiseringsdato': 'next_publication_date',
    'fyllingsgrad_forrige_uke': 'filling_degree_prev_week',
    'endring_fyllingsgrad': 'change_filling_degree'
}

df.rename(columns=column_mapping, inplace=True)

#Convert 'date' column to datetime format for easier manipulation and analysis
df['date'] = pd.to_datetime(df['date'])

# Sort the DataFrame by date to ensure chronological order
df.sort_values('date', inplace=True)
df.reset_index(drop=True, inplace=True)
```

In [16]:
```python
# Skriv ut sammendrag av datakontrakt og informasjon om dataframe
print("--- DATAFRAME INFO ---")
df.info()

print("\n--- STATISTISK OVERSIKT (NUMERISKE FELT) ---")
display(df.describe())

print("\n--- DE FØRSTE 5 RADENE ---")
display(df.head())
```

```
--- DATAFRAME INFO ---
<class 'pandas.DataFrame'>
RangeIndex: 14877 entries, 0 to 14876
Data columns (total 11 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   date                      14877 non-null  datetime64[us]
 1   area_type                 14877 non-null  str
 2   area_number               14877 non-null  int64
 3   iso_year                  14877 non-null  int64
 4   iso_week                  14877 non-null  int64
 5   filling_degree            14877 non-null  float64
 6   capacity_twh              14877 non-null  float64
 7   filling_twh               14877 non-null  float64
 8   next_publication_date     14877 non-null  str
 9   filling_degree_prev_week  14877 non-null  float64
 10  change_filling_degree     14877 non-null  float64
dtypes: datetime64[us](1), float64(5), int64(3), str(2)
memory usage: 1.6 MB

--- STATISTISK OVERSIKT (NUMERISKE FELT) ---
```

	date	area_number	iso_year	iso_week	filling_degree	capacity_twh	filling_twh	filling_degree_prev_week	change_filling_degree
count	14877	14877.000000	14877.000000	14877.000000	14877.000000	14877.000000	14877.000000	14877.000000	14877.000000
mean	2010-11-07 00:00:00	2.333333	2010.346038	26.405929	0.618852	29.145860	18.139713	0.618890	-0.000038
min	1995-01-08 00:00:00	0.000000	1995.000000	1.000000	0.055160	6.003264	0.331142	0.055160	-0.242484
25%	2002-12-08 00:00:00	1.000000	2002.000000	13.000000	0.456234	17.390587	7.321973	0.456356	-0.022395
50%	2010-11-07 00:00:00	2.000000	2010.000000	26.000000	0.648049	23.237715	14.501389	0.648114	-0.007072
75%	2018-10-07 00:00:00	3.000000	2018.000000	39.000000	0.802135	34.043590	22.203530	0.802133	0.016277
max	2026-09-06 00:00:00	5.000000	2026.000000	53.000000	1.020072	87.437580	84.544820	1.020037	0.336623
std	NaN	1.490762	9.146696	15.017910	0.214850	22.739070	15.966211	0.214843	0.031385

```
--- DE FØRSTE 5 RADENE ---
```

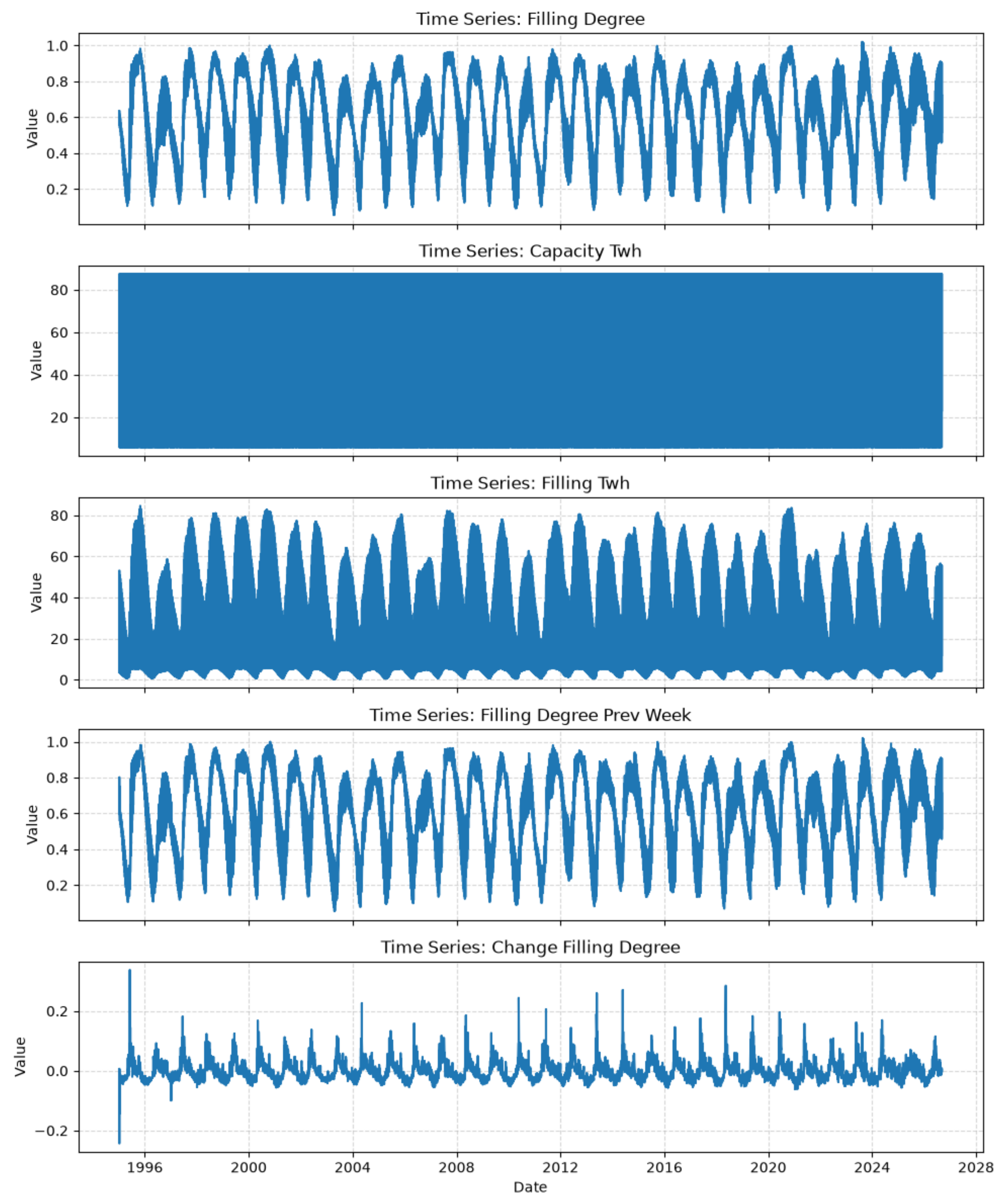
	date	area_type	area_number	iso_year	iso_week	filling_degree	capacity_twh	filling_twh	next_publication_date	filling_degree_prev_week	change_filling_degree
0	1995-01-08	EL	5	1995	1	0.616448	17.390587	10.720388	0001-01-01T00:00:00	0.614288	0.002160
1	1995-01-08	EL	4	1995	1	0.558864	21.079208	11.780413	0001-01-01T00:00:00	0.801348	-0.242484
2	1995-01-08	EL	3	1995	1	0.602376	8.920932	5.373754	0001-01-01T00:00:00	0.596027	0.006349
3	1995-01-08	EL	1	1995	1	0.621467	6.003264	3.730831	0001-01-01T00:00:00	0.734888	-0.113421
4	1995-01-08	EL	2	1995	1	0.630947	34.043590	21.479713	0001-01-01T00:00:00	0.777604	-0.146657

In [17]:
```python
# Velg ut numeriske kolonner som gir mening å plotte over tid
numeric_cols = ['filling_degree', 'capacity_twh', 'filling_twh', 'filling_degree_prev_week', 'change_filling_degree']

fig, axes = plt.subplots(len(numeric_cols), 1, figsize=(10, 12), sharex=True)

for ax, col in zip(axes, numeric_cols):
    ax.plot(df['date'], df[col], label=col, color='tab:blue')
    ax.set_title(f'Time Series: {col.replace("_", " ").title()}')
    ax.set_ylabel('Value')
    ax.grid(True, linestyle='--', alpha=0.5)

axes[-1].set_xlabel('Date')
plt.tight_layout()
plt.show()
```



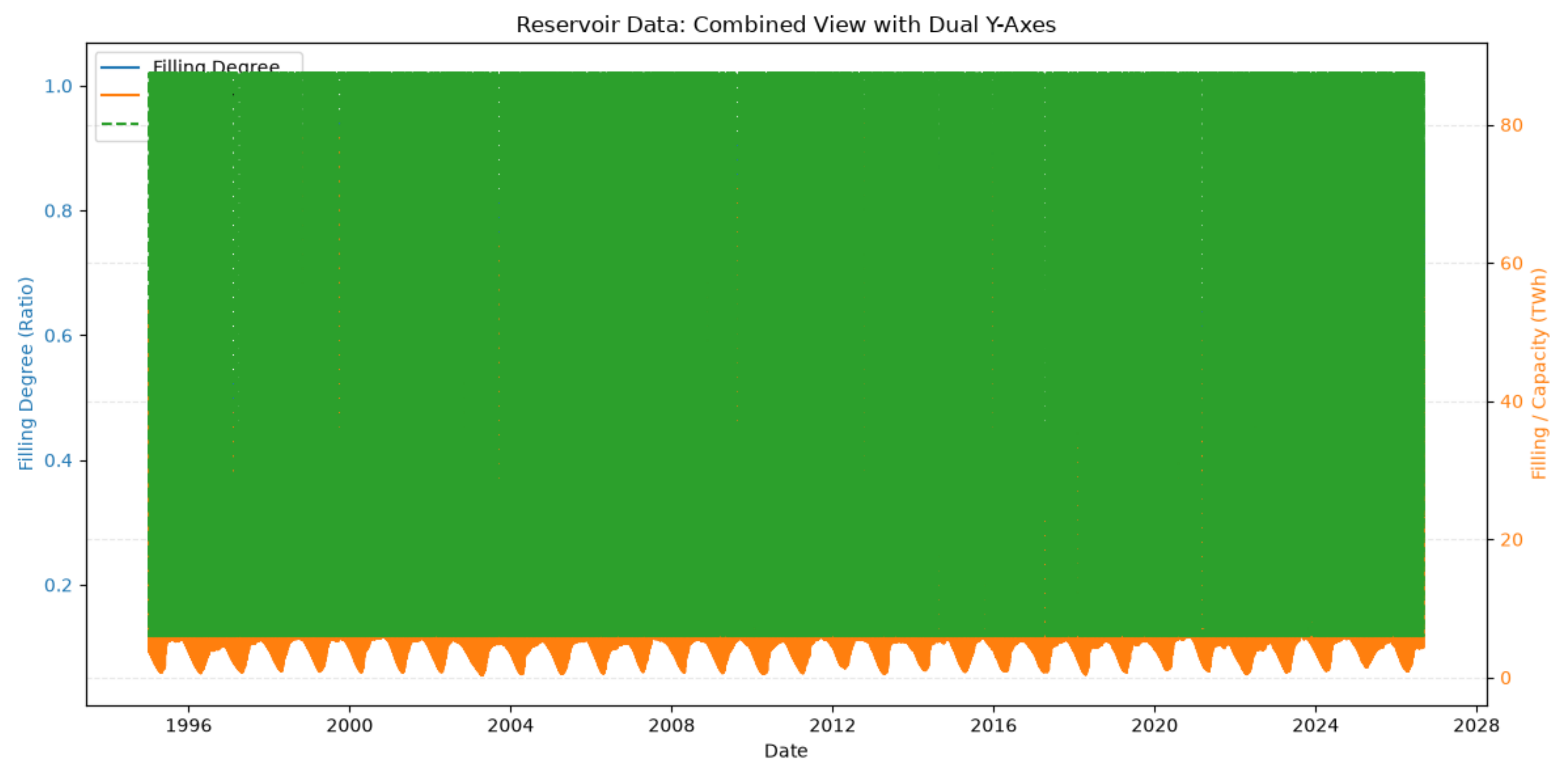
In [18]:
```python
fig, ax1 = plt.subplots(figsize=(12, 6))

# Aksesystem 1: Fyllingsgrad (0 til 1 scale)
color = 'tab:blue'
ax1.set_xlabel('Date')
ax1.set_ylabel('Filling Degree (Ratio)', color=color)
line1 = ax1.plot(df['date'], df['filling_degree'], color=color, label='Filling Degree')
ax1.tick_params(axis='y', labelcolor=color)

# Aksesystem 2: TWh (Større skala)
ax2 = ax1.twinx()
color = 'tab:orange'
ax2.set_ylabel('Filling / Capacity (TWh)', color=color)
line2 = ax2.plot(df['date'], df['filling_twh'], color=color, label='Filling (TWh)')
line3 = ax2.plot(df['date'], df['capacity_twh'], color='tab:green', linestyle='--', label='Capacity (TWh)')
ax2.tick_params(axis='y', labelcolor=color)

# Kombiner forklaringsskilt (legend) fra begge akser
lines = line1 + line2 + line3
labels = [l.get_label() for l in lines]
ax1.legend(lines, labels, loc='upper left')

plt.title('Reservoir Data: Combined View with Dual Y-Axes')
plt.grid(True, linestyle='--', alpha=0.3)
plt.tight_layout()
plt.show()
```



Log and AI Usage Statement AI Usage Description In this first assignment, AI tools (such as copilot/Gemini) were utilized as interactive programming assistants and writing sounding boards. Specifically, AI was used for: Translating structural templates and instructions into properly formatted Markdown and Jupyter Notebook code cells. Adapting previous template examples and boilerplate code to fit the specific context of the assigned housing unit/dataset. Drafting and refining English text sections, including the AI usage statement and the project work log. Guidance on establishing a clean workflow and repository structure for Python, Jupyter Notebook, and Streamlit integration.

The first phase of the IND320 project focused on setting up a reproducible development environment and creating an initial data analysis pipeline for the residential unit dataset. The primary goal was to establish a seamlessly integrated workflow spanning local development in Jupyter Notebook, version control via GitHub, and live web deployment through Streamlit.

The work commenced with setting up the project infrastructure. A dedicated public GitHub repository was created to serve as the core codebase and version control hub. Subsequently, a Streamlit account was linked to the GitHub repository to enable continuous deployment. A minimal working Streamlit application was created, committed, and deployed to ensure the hosting pipeline was fully functional before integrating complex data logic. Once the deployment pipeline was established, the focus shifted to local data processing and exploratory data analysis (EDA) within Jupyter Notebook. The dataset, containing time-series observations relevant to the residential unit, was loaded using Pandas with automatic datetime index parsing. Initial data verification involved examining dataset metadata, column data types, missing values, and descriptive statistics using methods such as .info() and .describe().T. This ensured the overall data quality before generating visualizations.

Data visualization was carried out in two main steps. First, each parameter within the dataset was plotted individually across time. This single-variable approach made it easier to identify individual signal behavior, missing intervals, and general time trends across environmental or energy-related metrics. Second, combined time-series plots were developed using Matplotlib with dual/split Y-axes (twinx()). Combining variables with different scales on a unified time axis allowed for direct visual comparison between key parameters (such as comparing outdoor weather variables against indoor environment or energy metrics).

Finally, documentation was integrated directly into the notebook. Clear Markdown headings, explanatory comments for code blocks, a summary of AI tool usage, and a detailed project log were added to ensure full transparency and reproducibility. Overall, Part 1 successfully established both the technical infrastructure and the baseline analytical insights needed for subsequent project assignments.